

Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester IV)

DATA VISUALIZATION WITH POWERBI AND TABLEAU

Practical – IX

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Date of Assignment: 27/08/2026	Date/Time of Submission:27/08/2026

Aim : Cleaning and Structuring Data for Analysis**A) Identify and handle missing or inconsistent data:****Step 1:** Load the Dataset

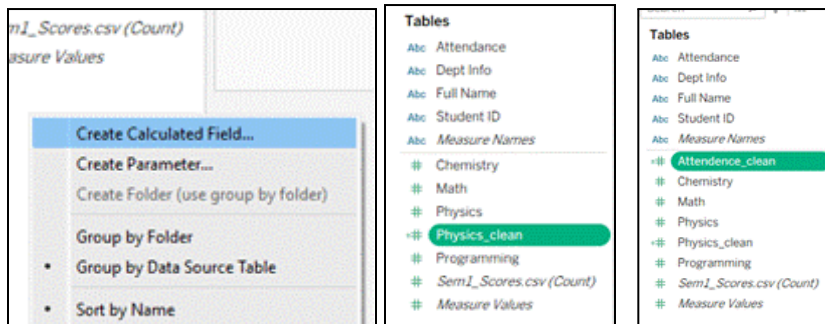
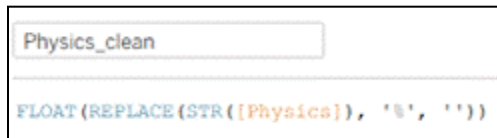
- . Open Tableau Desktop and load the given dataset.
- . Go to Sheet 1.

Step 2: Clean Physics and Attendance

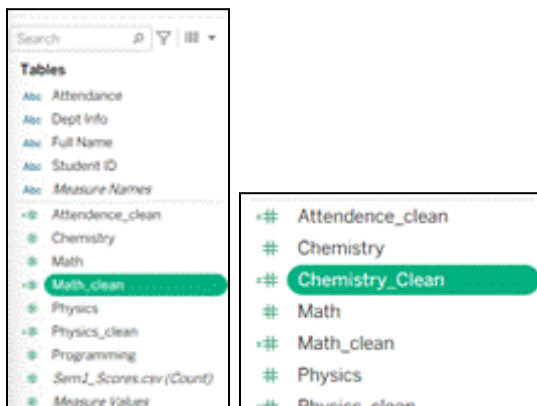
- . Right-click in the Data Pane - Create Calculated Field.
- . Create Physics_Clean.

Formula: `FLOAT (REPLACE (STR( [Physics] ), '%', ''))`

- . Repeat the same steps and create Attendance_Clean.

Formula : `FLOAT (TRIM (REPLACE (STR ( [Attendance] ), '%', '')))`**Step 3:** Handle Missing Values

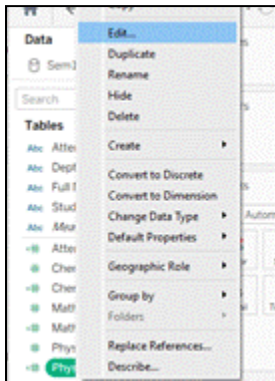
- . Create Math_Clean using `IFNULL([Math], 0)`.
- . Create Chemistry_Clean using `IFNULL([Chemistry], 0)`.



Step 4: Handle Missing Physics Values

. Right-click Physics_Clean - Edit and use:

Formula: IFNULL(FLOAT (REPLACE (STR([Physics]), '%', '')), 0)

**B) Convert Wide Data into Tall Format:**

Step 1: Open Data Source . Open the Data Source tab.

Step 2: Pivot Columns:

- . Hold Ctrl and select Math, Physics, Chemistry, Programming.
- . Right-click the selected columns - Pivot.

Math	▼	Create Calculated Field...
Physics	▼	Pivot
Chemistry	▼	Merge Mismatched Fields
Programming	▼	

Field Name	Physical Table
Dept Info	Sem1_Score
Attendance	Sem1_Score
Pivot Field Names	Pivot
Pivot Field Values	Pivot
Physics_clean	Calculation
Attendance_clean	Calculation
Math_clean	Calculation

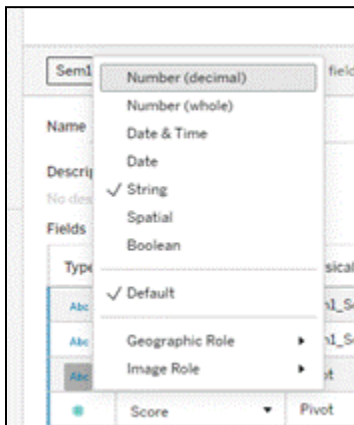
Step 3: Rename Pivoted Fields

- . Rename Pivot Field Names to Subject.
- . Rename Pivot Field Values to Score.

Attendance	▼	S
Subject	▼	P
Score	▼	P
Physics_clean	▼	C
Attendance_clean	▼	C

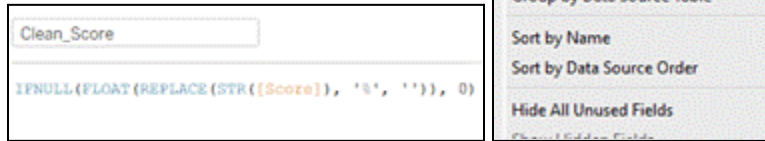
Step 4: Format Score:. Click the Abc icon above Score.

- . Select Number (decimal).



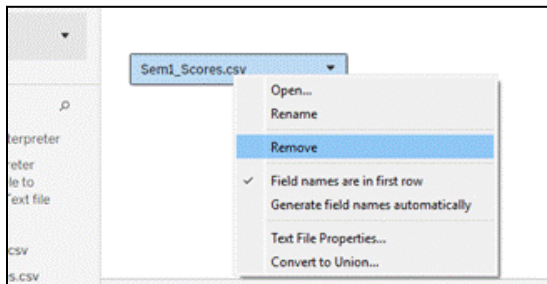
Alternative: Create Clean_Score

Formula: IFNULL (FLOAT (REPLACE (STR([Score]), '%', '')), 0)



C) Perform Union of Multiple Files:

Step 1: Open the Data: Clear the existing canvas



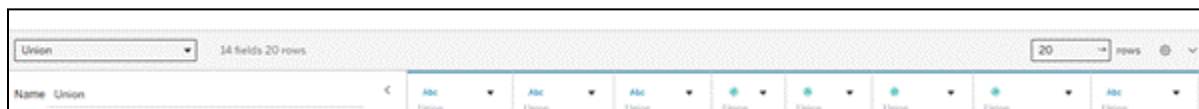
Step 2: Create a Multi-File Union

- . Drag New Union to the canvas.
- . Add Sem1_Scores.csv and Sem2_Scores.csv.



Step 3: Verify the Merged Data Grid

- . Check the data grid at the bottom.
- . Confirm that the total is 20 rows.
- . Check the Table Name column for the two files.



3. Scroll to the far right of the grid to verify the generated system column: Table Name: Displays "Sem1_Scores.csv" for the first 10 rows and "Sem2_Scores.csv" for the remaining 10 rows.

	Abc Union	Calculation	Calculation
Attendance	Table Name	Physics_clean	Attendance
%	Sem1_Scores.csv	0.900000	
%	Sem1_Scores.csv	0.750000	
%	Sem1_Scores.csv	0.820000	
%	Sem1_Scores.csv	0.880000	
%	Sem1_Scores.csv	0.000000	
%	Sem1_Scores.csv	0.910000	
%	Sem1_Scores.csv	0.700000	

Step 4: Create a Clean Semester Identifier Dimension

1. Click on Sheet 1 at the bottom-left.
2. Create Calculated Field. Set the Field Name to: Semester
4. Enter the following formula:

Create Calculated Field...

Create Parameter...

Create Folder (use group by folder)

Group by Folder

Group by Data Source Table

Semester

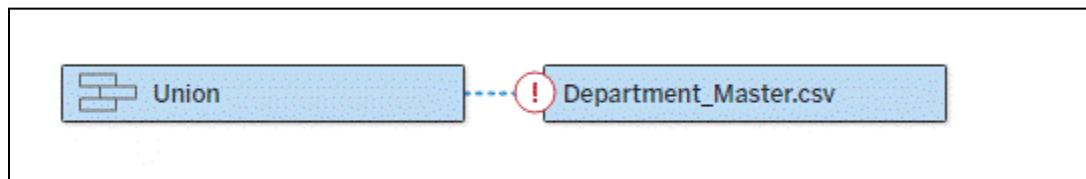
```
IF CONTAINS([Table Name], "Sem1") THEN "Semester 1"
ELSEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2"
ELSE "Other"
END
```

D) Execute cross-database joins:

Step 1: Stay on the Top Canvas. Leave your Union table box as it is on the canvas.

Step 2: Add and Link Department_Master.csv

1. In the left sidebar under Files / Connections, find Department_Master.csv. 2. Drag Department_Master.csv onto the canvas and drop it to the right of your Union box.
3. A flexible line (relationship / join noodle) will connect the two boxes.



4. If the relationship dialog opens (or click the relationship line):
 - o Left Table Field (Union): Select Dept_Info
 - o Right Table Field (Department_Master): Select Dept_Code
 - o Operator: =

Union — Department...

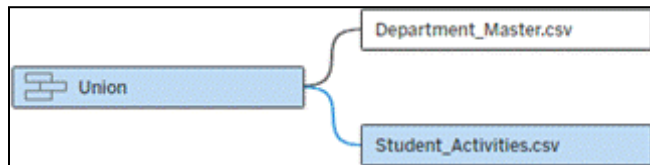
Name: Department_Master.csv

How do relationships differ from joins? [Learn more](#)

Union	Operator	Department_Maste...
Abc Dept_Info	=	Abc Dept Code

Step 3: Add and Link Student_Activities.csv

1. From the left sidebar, drag Student_Activities.csv onto the canvas.
2. Drop it to the right of the Union box.



3. In the relationship dialog:
 - o Left Table Field (Union): Select Student_ID
 - o Right Table Field (Student_Activities): Select Student_ID
 - o Operator: =

Union — Student_A...

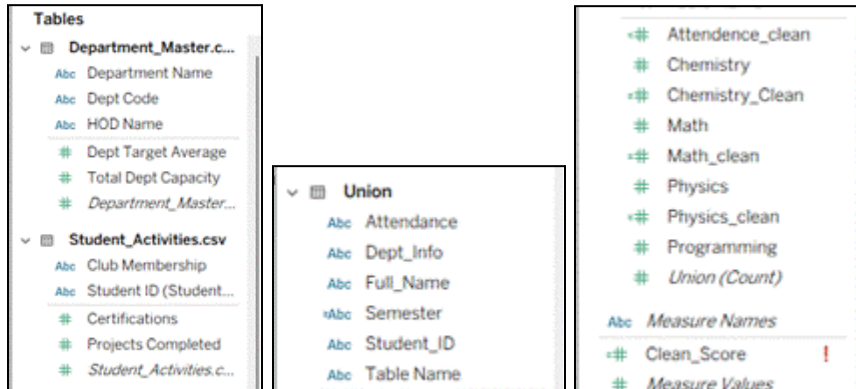
Name: Student_Activities.csv

How do relationships differ from joins? [Learn more](#)

Union	Operator	Student_Activities....
Abc Student_ID	=	Abc Student ID (Stu...

Step 4: Verify in Sheet 1

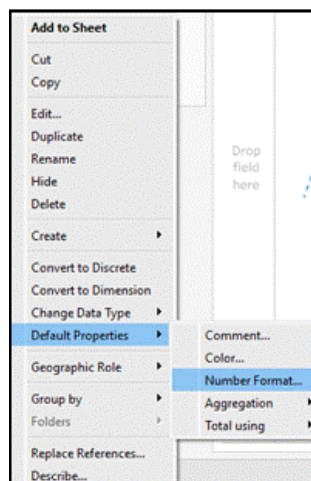
1. Click on Sheet 1 at the bottom-left.
2. In the Data Pane on the far left, you will now see all three tables organized with their respective fields:
 - o Union: Student_ID, Full_Name, Dept_Info, Math, Physics, Chemistry, Programming, Attendance, Table Name
 - o Department_Master: Dept_Code, Department_Name, HOD_Name, Dept_Target_Average, Total_Dept_Capacity
 - o Student_Activities: Club_Membership, Projects_Completed, Certifications

**E.) Restructure Data for Better Visualization:****Step 1: Calculate Total Marks**

1. Open Tableau.
2. Go to Sheet 1.
3. In the Data Pane, right-click on an empty area.
4. Select Create Calculated Field.
5. Enter field name: Total_Score
6. Enter formula: $[\text{Math_Clean}] + [\text{Physics_Clean}] + [\text{Chemistry_Clean}] + [\text{Programming}]$
7. Click OK.

Step 2: Calculate Overall Percentage

1. Right-click in the Data Pane.
2. Select Create Calculated Field.
3. Field name: Overall_Percentage
4. Formula: $[\text{Total_Score}] / 400.0$
5. Click OK.
6. Right-click Overall_Percentage.
7. Select Default Properties → Number Format.
8. Select Percentage.
9. Set decimal places to 1.
10. Click OK.



Step 3: Create Performance Band

1. Right-click in Data Pane.
2. Select Create Calculated Field.
3. Field name: Performance_Band

Enter:

```
IF [Overall_Percentage] >= 0.85 THEN "Distinction"
ELSEIF [Overall_Percentage] >= 0.75 THEN "First Class"
ELSEIF [Overall_Percentage] >= 0.60 THEN "Second Class"
ELSE "Needs Academic Support"
4. END
5. Click OK.
```

Step 4: Create Attendance Status

1. Right-click Data Pane.
2. Select Create Calculated Field.
3. Field name: Attendance_Status

Formula: IF [Attendance_clean] >= 75 THEN
"Eligible (Good Standing)"

ELSE

"Shortage Warning (<75%)"

END

4. END
5. Click OK.

Step 5: Build Student Performance Scorecard

Now create the visualization:

1. Drag Student_ID → Rows.
2. Drag Full_Name → Rows.
3. Drag Semester → Columns.
4. Drag Overall_Percentage → Text in Marks card.
5. Drag Performance_Band → Color.
6. Drag Attendance_Status → Tooltip or Detail.

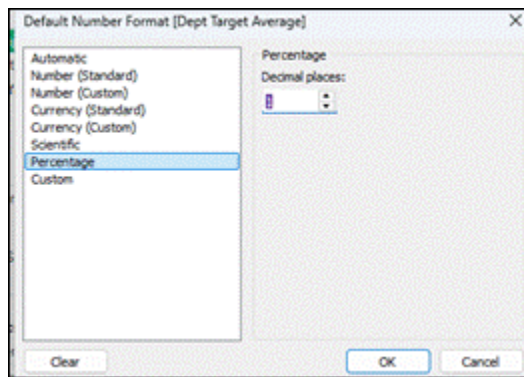
Output:-

Student ID	Full Name	Semester
S101	Aarav Sharma	67.2%
S102	priya VERMA	
S103	Rohan Gupta	69.7%
S104	Sneha Patel	
S105	aditya Rao	66.5%
S106	Ananya Singh	
S107	Rahul Nair	71.5%
S108	Kavya Reddy	49.4%
S109	Vikram Joshi	63.0%
S110	Pooja Iyer	

F. Handle Different Levels of Detail Using LOD

Step 1: Department Average

1. Go to Sheet 1 or create Sheet 2.
2. Right-click in Data Pane.
3. Select Create Calculated Field.
4. Field name: Dept_Average_Percentage_LOD
5. Formula: { FIXED [Dept_Info] : AVG([Overall_Percentage]) }
6. Click OK.
7. Right-click the new field.
8. Select Default Properties → Number Format.
9. Select Percentage and 1 decimal place.



Step 2: Student Career Average

1. Right-click Data Pane.
2. Select Create Calculated Field.
3. Field name:
Student_Overall_Career_Average
4. Formula:
{ FIXED [Student_ID] : AVG([Overall_Percentage]) }
5. Click OK.
6. Set the number format to Percentage with 1 decimal place.

Step 3: Calculate Variance

1. Create another Calculated Field.
2. Field name:
Variance_From_Dept_Average
3. Formula:
[Overall_Percentage] - [Dept_Average_Percentage_LOD]
4. Click OK.
5. Set it to Percentage with 1 decimal place.

Step 4: Create Benchmark Status

1. Create another Calculated Field.
2. Field name: Benchmark_Status

Formula:

IF [Variance_From_Dept_Average] >= 0 THEN "Above Department Benchmark"
ELSE "Below Department Benchmark"

3. END
4. Click OK.

Step 5: Build LOD Visualization

1. Drag Dept_Info → Rows.
2. Drag Student_ID → Rows.
3. Drag Full_Name → Rows.
4. Drag Semester → Columns.
5. Drag Overall_Percentage → Text.
6. Drag Dept_Average_Percentage_LOD → Tooltip.
7. Drag Benchmark_Status → Color.

Final result

Dept Info	Student ID	Full Name	Semester	Overall_Percentage	Benchmark_Status
CSE-01	5101	Aarav Sharma	r 2	67.2%	Above Department Benchmark
	5103	Rohan Gupta		66.5%	
	5106	Ananya Singh		66.5%	
	5109	Vikram Joshi		66.5%	
ECE-02	5102	griya VERMA	r 2	63.0%	Below Department Benchmark
	5107	Rahul Kaur		63.0%	
	5110	Rishi Iyer		63.0%	
IT-03	5104	Sneha Patel	r 2	69.7%	Above Department Benchmark
	5108	Kavya Reddy		71.5%	
MECH-04	5105	aditya Rao	r 2	71.5%	Above Department Benchmark

G: Apply Advanced Data Cleaning Techniques**STEP 1 — Full_Name_Clean**

1. Sheet 1 par raho.
2. Left side Data Pane mein right-click karo.
3. Create Calculated Field select karo.
4. Field Name: Full_Name_Clean
5. Formula:

UPPER(LEFT(TRIM([Full_Name]), 1)) +

LOWER(MID(TRIM([Full_Name]), 2, FIND(TRIM([Full_Name]), " ") - 1)) +
" " +

UPPER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 1, 1)) +

LOWER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 2, LEN(TRIM([Full_Name]))))

6. OK click karo.

STEP 2 — First_Name

1. Data Pane → Right Click
2. Create Calculated Field
3. Name:
First_Name
4. Formula:

TRIM(SPLIT([Full_Name], " ", 1))

5. OK.

STEP 3 — Last_Name

1. Data Pane → Right Click
2. Create Calculated Field
3. Name:
Last_Name
4. Formula:

TRIM(SPLIT([Full_Name], " ", -1))

5. OK.

STEP 4 — Dept_Branch

Tumhare Dept_Info mein values jaise CSE-01, ECE-02 hain.

1. Create Calculated Field
2. Name:
Dept_Branch
3. Formula:

SPLIT([Dept_Info], "-", 1)

4. OK.

STEP 5 — Section_ID

1. Create Calculated Field
2. Name:
Section_ID
3. Formula:

SPLIT([Dept_Info], "-", 2)

4. OK.

STEP 6 — Student_Display_Label

1. Create Calculated Field
2. Name:
Student_Display_Label
3. Formula:

[Student_ID] + " - " + [Full_Name_Clean] + " (" + [Dept_Branch] + ") "

4. OK.